

Лабораторная работа №11

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

На сервере зададим пароль для пользователя root (рис. @fig-1):



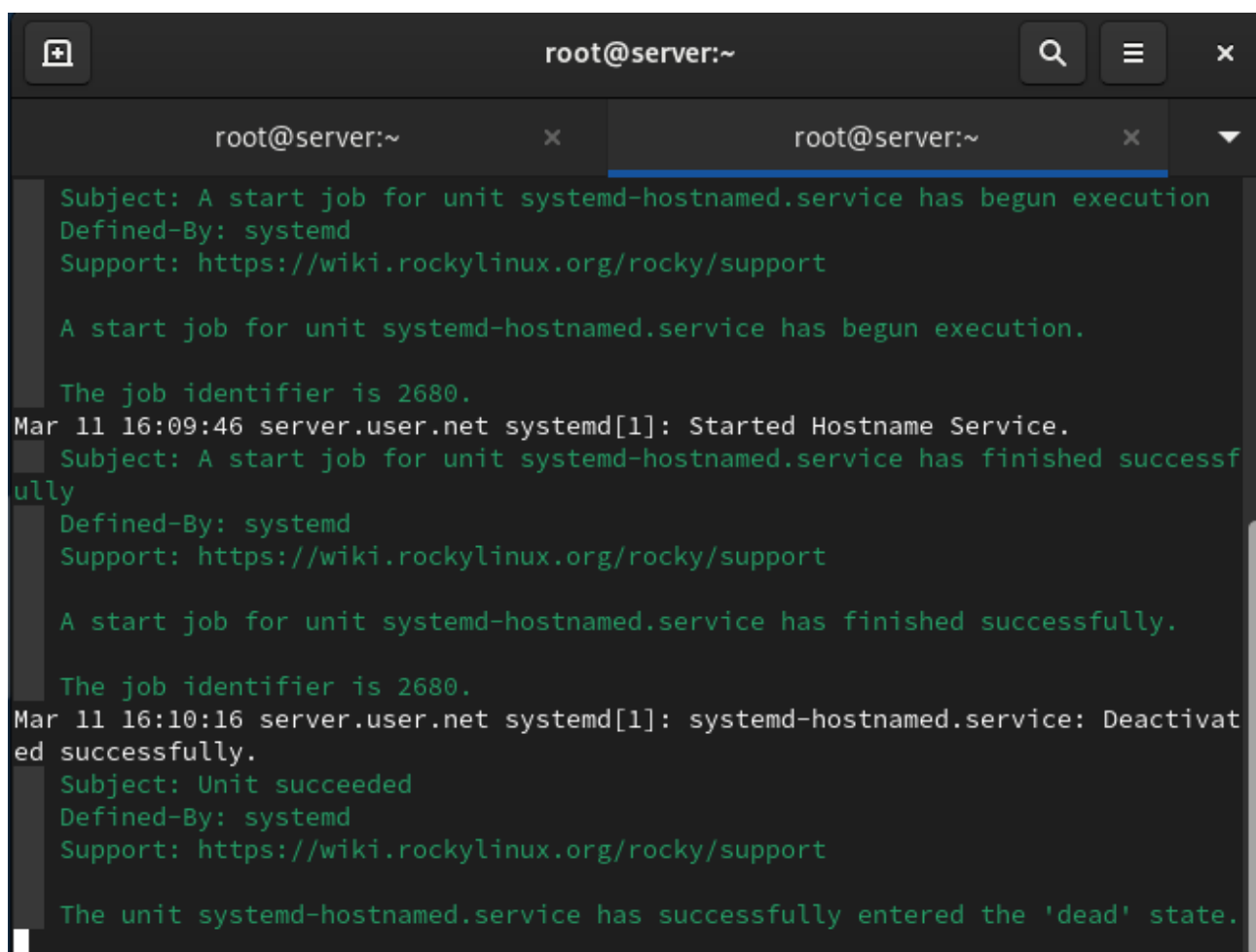
```
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]# passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]#
```

Рис. 1: Этап 1

На сервере в дополнительном терминале запустим мониторинг системных событий (рис. @fig-2):

С клиента попытаемся получить доступ к серверу посредством SSH-соединения через пользователя root (рис. @fig-3):

На сервере откроем файл /etc/ssh/sshd_config для редактирования и запретим вход на сервер пользователю root (рис. @fig-4):



The image shows a terminal window with a dark background. At the top, there's a title bar with a window icon, the text 'root@server:~', and search, menu, and close buttons. Below the title bar, there are two tabs, both labeled 'root@server:~'. The main area of the terminal displays a series of green-colored log messages from systemd. The messages indicate the start and successful completion of a job for the 'systemd-hostnamed.service' unit, with a job identifier of 2680. The logs are timestamped 'Mar 11 16:09:46' and 'Mar 11 16:10:16'. The messages include 'Subject: A start job for unit systemd-hostnamed.service has begun execution', 'Defined-By: systemd', 'Support: https://wiki.rockylinux.org/rocky/support', and 'The job identifier is 2680.' followed by 'Started Hostname Service.' and 'Deactivated successfully.'.

```
Subject: A start job for unit systemd-hostnamed.service has begun execution
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit systemd-hostnamed.service has begun execution.

The job identifier is 2680.
Mar 11 16:09:46 server.user.net systemd[1]: Started Hostname Service.
Subject: A start job for unit systemd-hostnamed.service has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit systemd-hostnamed.service has finished successfully.

The job identifier is 2680.
Mar 11 16:10:16 server.user.net systemd[1]: systemd-hostnamed.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit systemd-hostnamed.service has successfully entered the 'dead' state.
```

Рис. 2: Этап 2

После сохранения изменений перезапустим sshd (рис. @fig-5):

На сервере откроем файл `/etc/ssh/sshd_config` и добавим строку `AllowUsers vagrant`, затем перезапустим sshd (рис. @fig-6, @fig-7):

Настройка `AllowUsers`

Внесём изменение в файл `/etc/ssh/sshd_config`, добавив пользователя `user` в `AllowUsers`, и перезапустим sshd (рис. @fig-8, @fig-9):

Добавление пользователя в `AllowUsers`

В файле `/etc/ssh/sshd_config` добавим второй порт `2022`, перезапустим sshd и проверим статус (рис. @fig-10, @fig-11):

Настройка нескольких портов

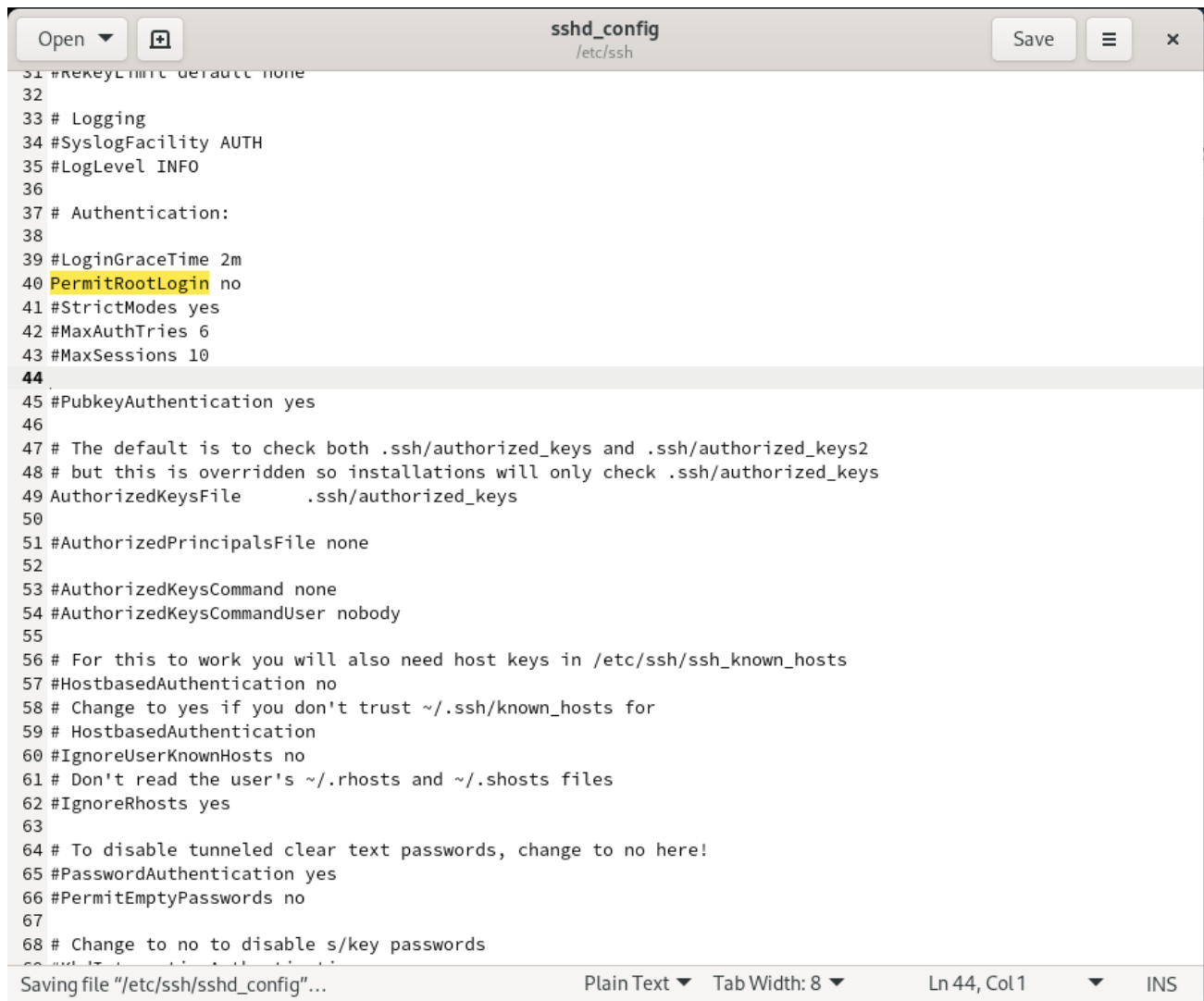


Рис. 3: Этап 3

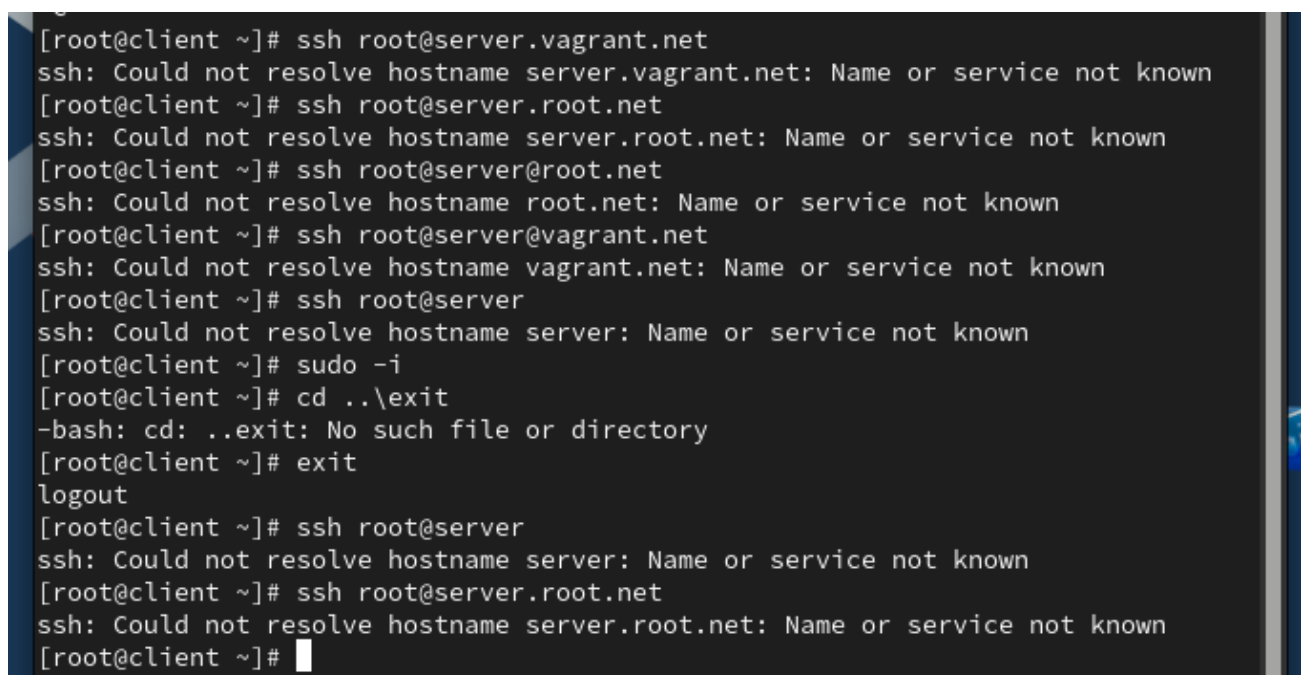


Рис. 4: Этап 4

```

ssh: Could not resolve hostname server: Name or service not known
[root@client ~]# ssh root@server.root.net
ssh: Could not resolve hostname server.root.net: Name or service not known
[root@client ~]# ssh vagrant@server.vagrant.net
ssh: Could not resolve hostname server.vagrant.net: Name or service not known
[root@client ~]# ssh vagrant@server.vagrant.net
ssh: Could not resolve hostname server.vagrant.net: Name or service not known
[root@client ~]#

```

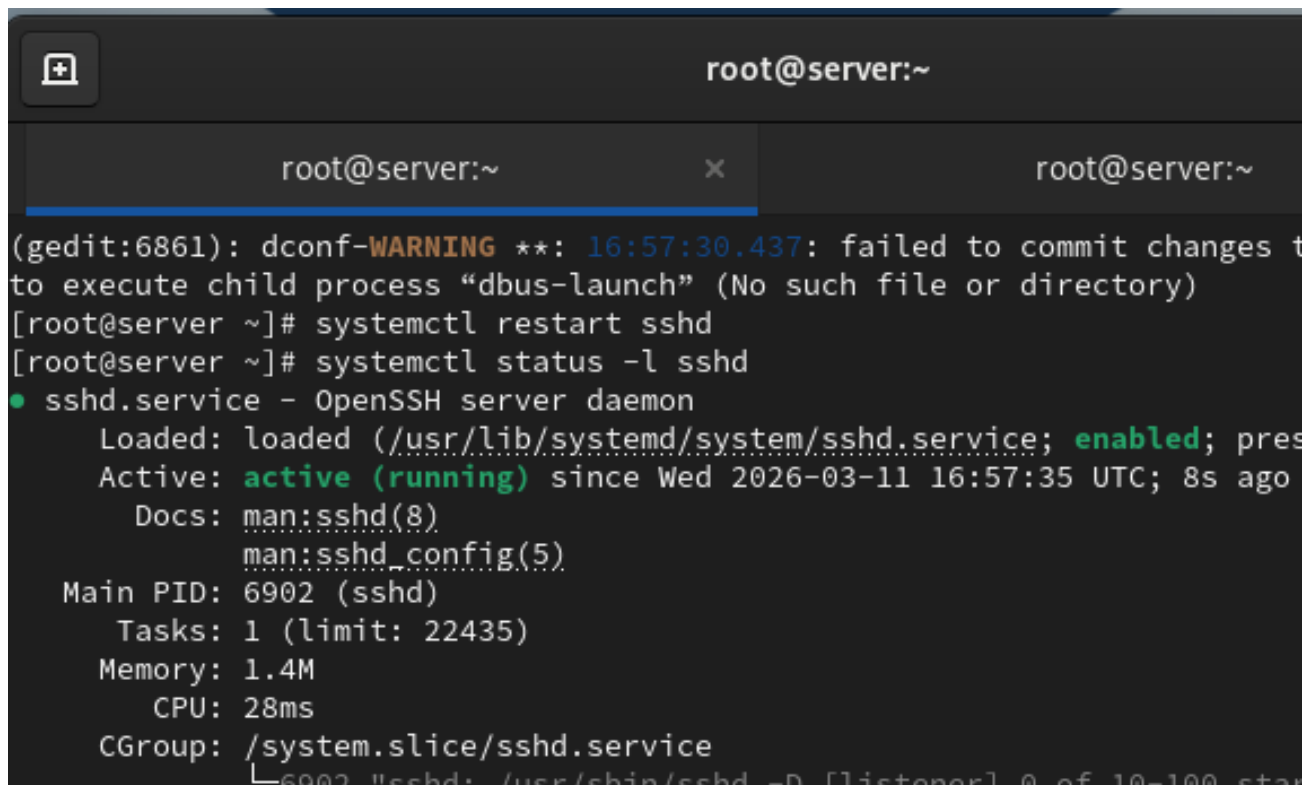
Рис. 5: Этап 5

```

93 # and KbdInteractiveAuthentication to 'no'.
94 # WARNING: 'UsePAM no' is not supported in RHEL and may cause several
95 # problems.
96 #UsePAM no
97
98 #AllowAgentForwarding yes
99 #AllowTcpForwarding yes
100 #GatewayPorts no
101 #X11Forwarding no
102 #X11DisplayOffset 10
103 #X11UseLocalhost yes
104 #PermitTTY yes
105 #PrintMotd yes
106 #PrintLastLog yes
107 #TCPKeepAlive yes
108 #PermitUserEnvironment no
109 #Compression delayed
110 #ClientAliveInterval 0
111 #ClientAliveCountMax 3
112 #UseDNS no
113 #PidFile /var/run/sshd.pid
114 #MaxStartups 10:30:100
115 #PermitTunnel no
116 #ChrootDirectory none
117 #VersionAddendum none
118
119 # no default banner path
120 #Banner none
121
122     AllowUsers vagrant user
123
124 # override default of no subsystems
125 Subsystem      sftp    /usr/libexec/openssh/sftp-server
126
127 # Example of overriding settings on a per-user basis
128 #Match User anoncvs
129 #       X11Forwarding no
130 #       AllowTcpForwarding no

```

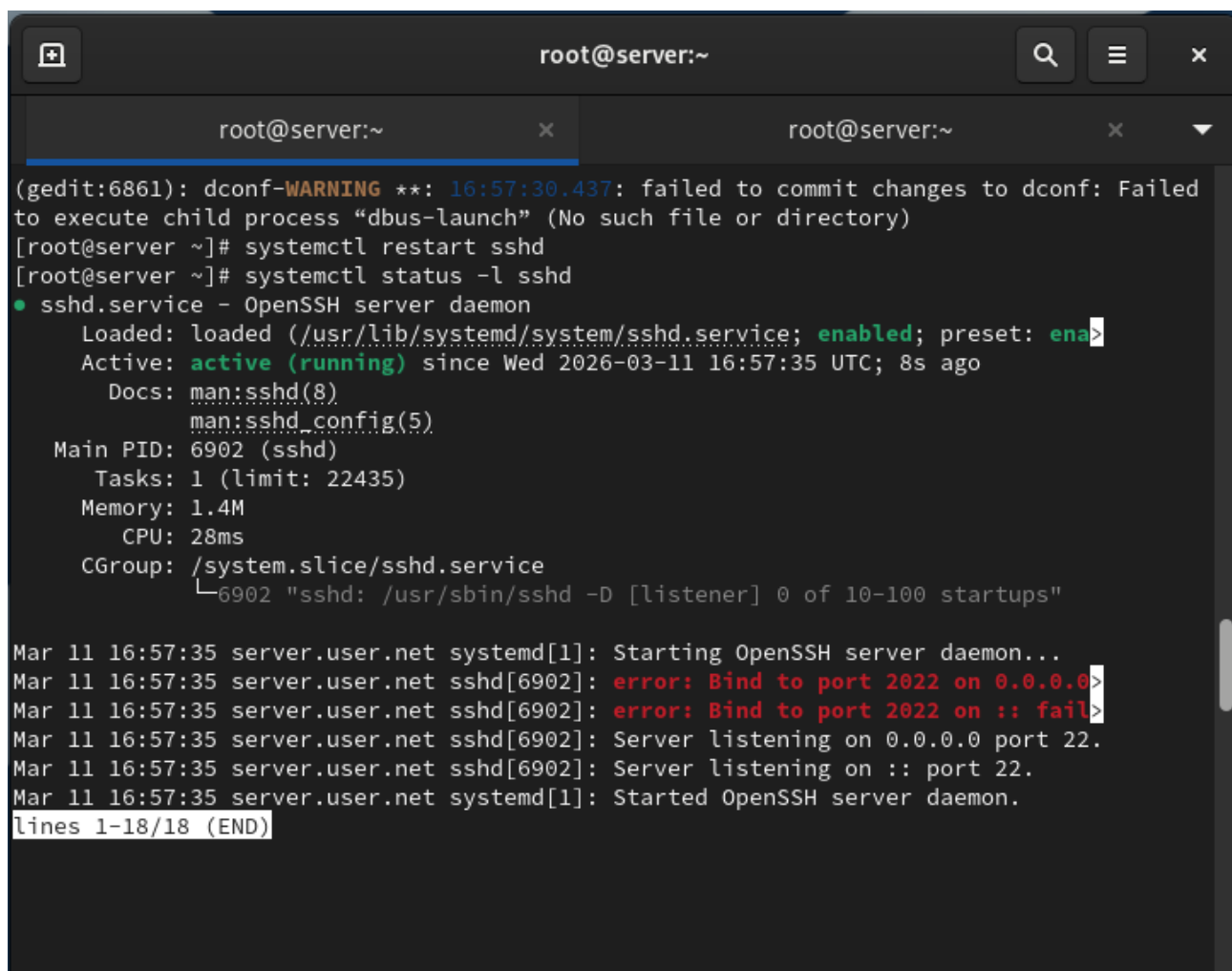
Рис. 6: Этап 6



A terminal window titled 'root@server:~' with two tabs. The active tab shows the output of 'systemctl status -l sshd'. The output indicates that the 'sshd.service' is loaded and active (running) since Wed 2026-03-11 16:57:35 UTC. It lists the main PID as 6902, tasks as 1, memory as 1.4M, and CPU as 28ms. The CGroup is '/system.slice/sshd.service'. A warning message from gedit is visible at the top: '(gedit:6861): dconf-WARNING **: 16:57:30.437: failed to commit changes to execute child process "dbus-launch" (No such file or directory)'.

```
root@server:~  
(gedit:6861): dconf-WARNING **: 16:57:30.437: failed to commit changes to execute child process "dbus-launch" (No such file or directory)  
[root@server ~]# systemctl restart sshd  
[root@server ~]# systemctl status -l sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)  
   Active: active (running) since Wed 2026-03-11 16:57:35 UTC; 8s ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Main PID: 6902 (sshd)  
    Tasks: 1 (limit: 22435)  
  Memory: 1.4M  
     CPU: 28ms  
   CGroup: /system.slice/sshd.service  
           └─6902 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Рис. 7: Этап 7



A terminal window titled 'root@server:~' with two tabs. The active tab shows the output of 'systemctl status -l sshd' and a series of logs. The status output is identical to the previous image. Below it, the logs show the startup of the OpenSSH server daemon. The logs indicate an error: 'error: Bind to port 2022 on 0.0.0.0: error: Bind to port 2022 on :: fail'. The logs also show the server listening on 0.0.0.0 port 22 and on :: port 22. The logs are dated Mar 11 16:57:35. The terminal window has a search bar and a menu icon in the top right corner.

```
root@server:~  
(gedit:6861): dconf-WARNING **: 16:57:30.437: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)  
[root@server ~]# systemctl restart sshd  
[root@server ~]# systemctl status -l sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)  
   Active: active (running) since Wed 2026-03-11 16:57:35 UTC; 8s ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Main PID: 6902 (sshd)  
    Tasks: 1 (limit: 22435)  
  Memory: 1.4M  
     CPU: 28ms  
   CGroup: /system.slice/sshd.service  
           └─6902 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Mar 11 16:57:35 server.user.net systemd[1]: Starting OpenSSH server daemon...  
Mar 11 16:57:35 server.user.net sshd[6902]: error: Bind to port 2022 on 0.0.0.0: error: Bind to port 2022 on :: fail  
Mar 11 16:57:35 server.user.net sshd[6902]: Server listening on 0.0.0.0 port 22.  
Mar 11 16:57:35 server.user.net sshd[6902]: Server listening on :: port 22.  
Mar 11 16:57:35 server.user.net systemd[1]: Started OpenSSH server daemon.  
Lines 1-18/18 (END)
```

Рис. 8: Этап 8

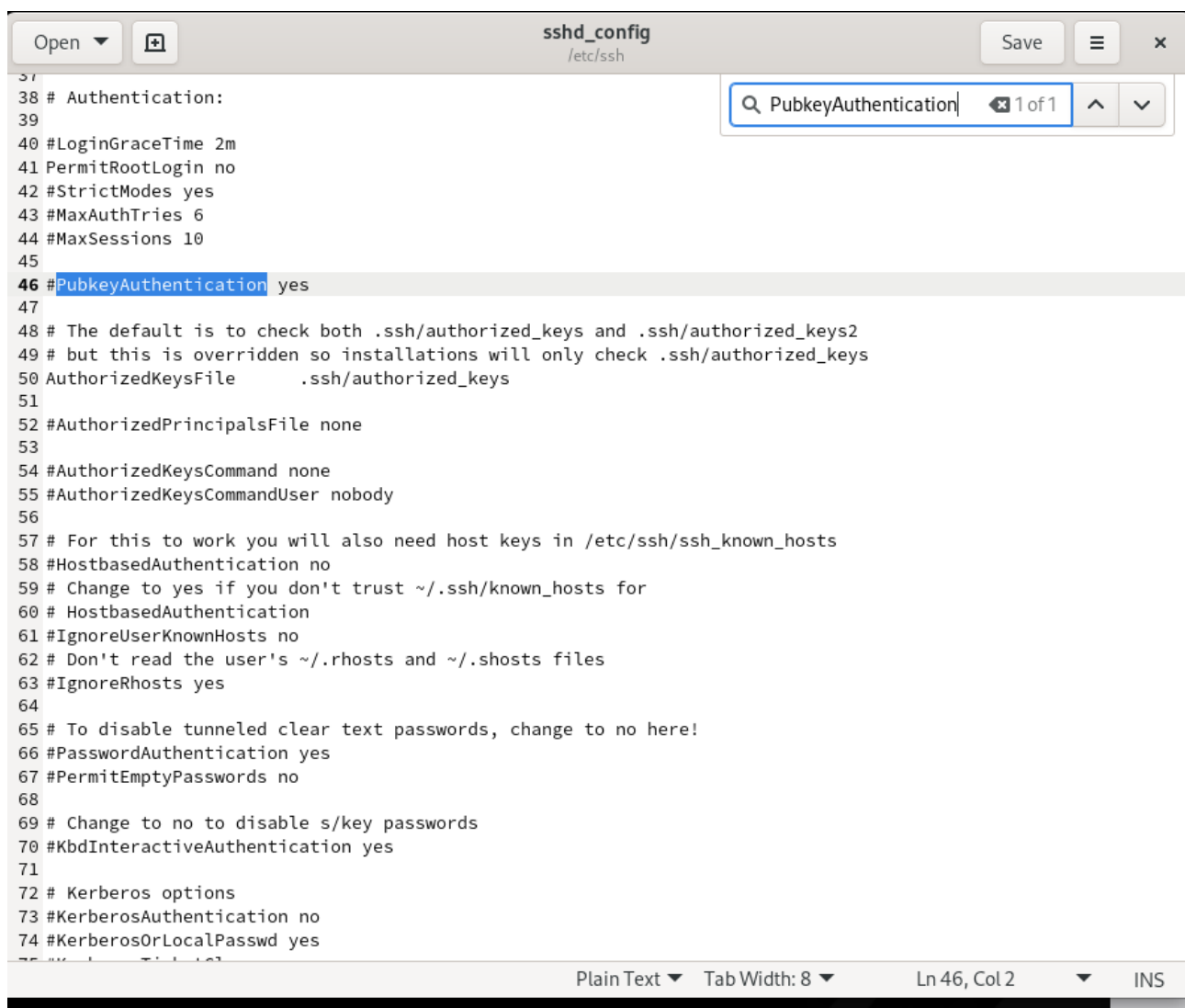


Рис. 9: Этап 9

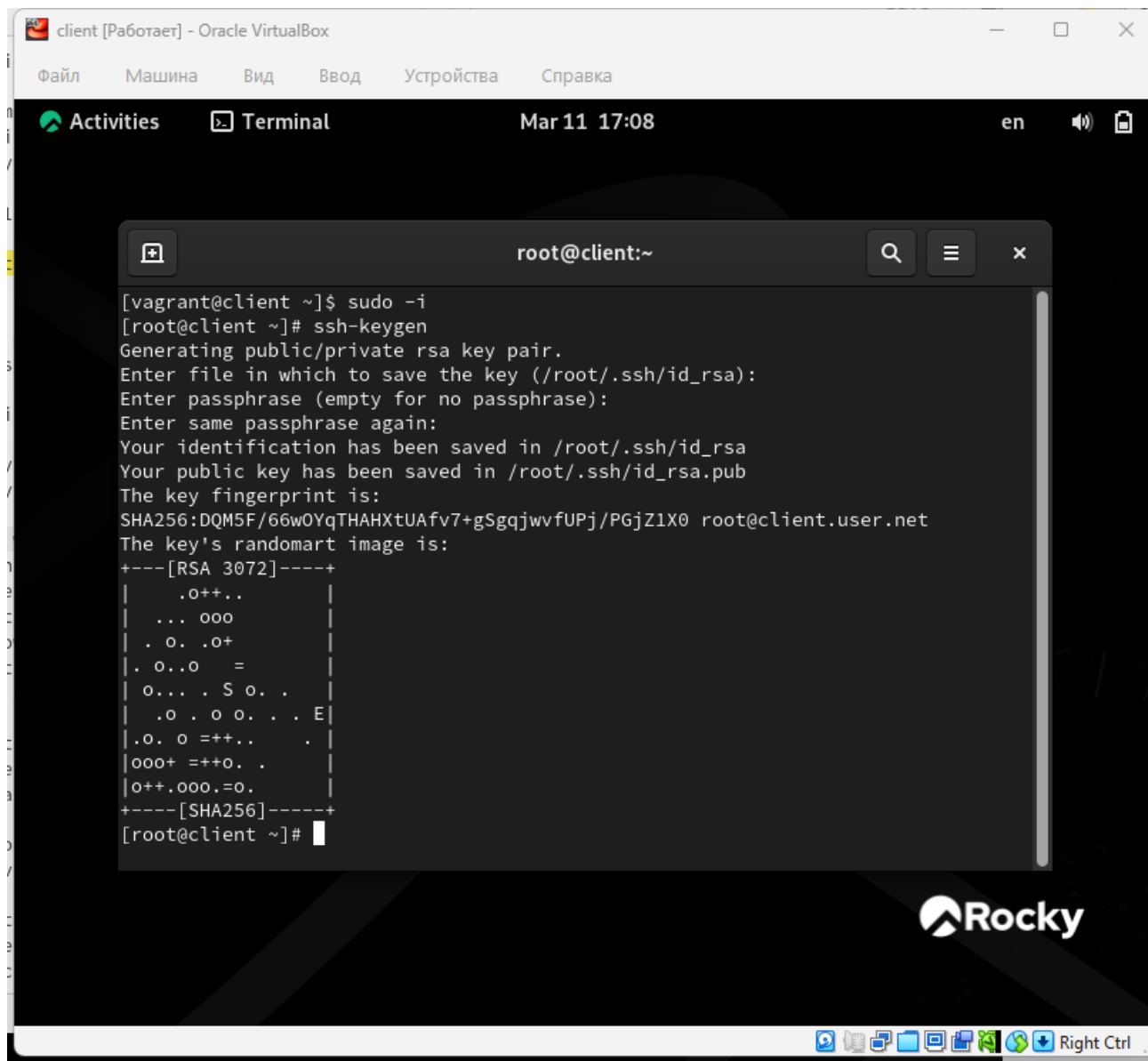


Рис. 10: Этап 10

```
vagrant@client:~
The key's randomart image is:
+---[RSA 3072]-----+
|  .o++..          |
| ... ooo          |
| . o. .o+         |
| . o..o   =       |
| o... . S o. .    |
| .o . o o. . . E  |
|.o. o =++..      .|
|ooo+ =++o. .     |
|o++..ooo.=o.     |
+-----[SHA256]-----+
[root@client ~]# ssh-copy-id user@server.user.net
/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out
any that are already installed
/bin/ssh-copy-id: ERROR: ssh: Could not resolve hostname server.user.net: Name or
service not known
[root@client ~]#
logout
[vagrant@client ~]$
```

Рис. 11: Этап 11

Исправим метки SELinux для порта 2022 и откроем порт в межсетевом экране, затем перезапустим sshd (рис. @fig-12):

С клиента подключимся к серверу через порт 2022 и получим доступ root (рис. @fig-13):

На сервере разрешим аутентификацию по ключу и перезапустим sshd (рис. @fig-14):

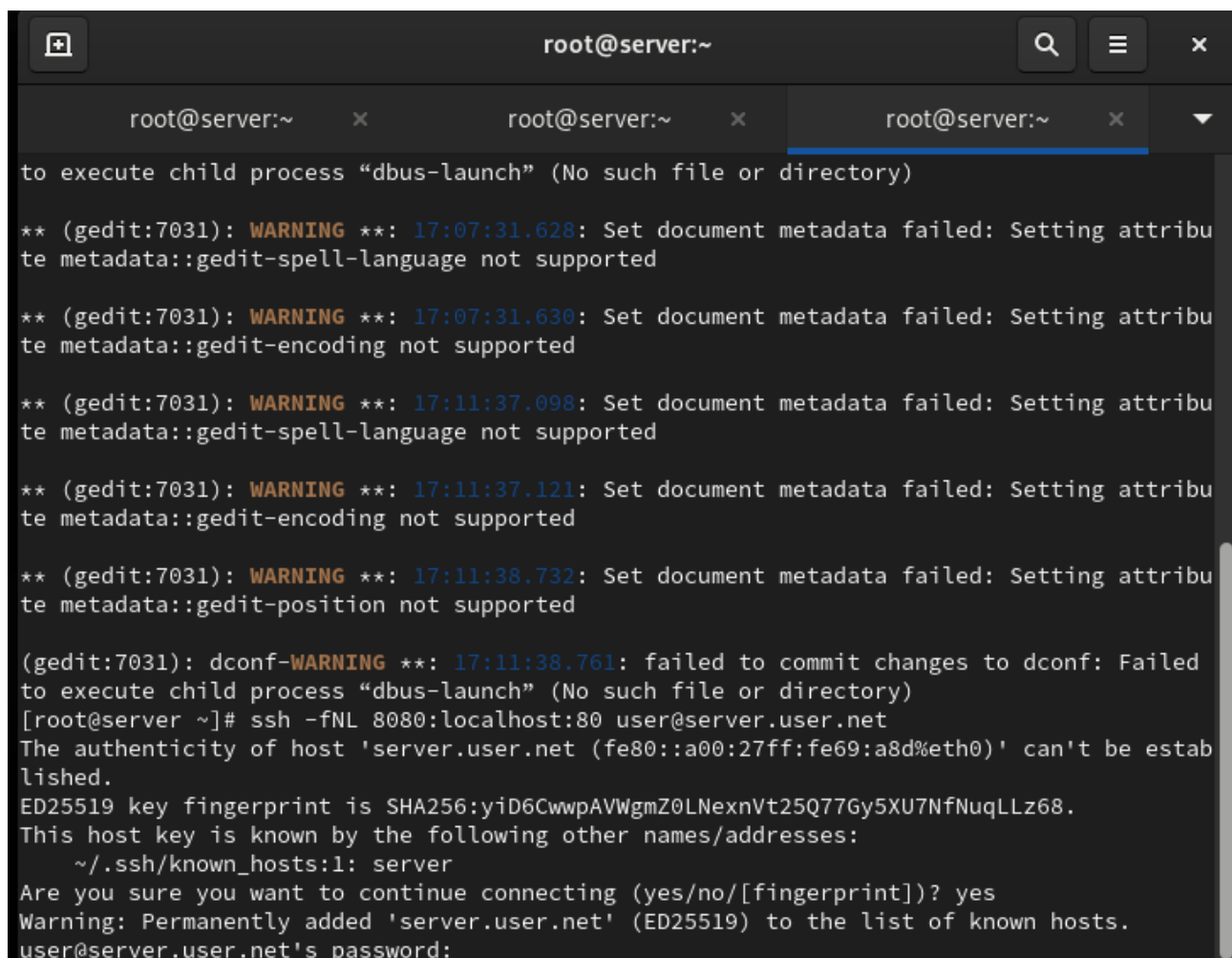
На клиенте сформируем SSH-ключ и скопируем его на сервер (рис. @fig-15):

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

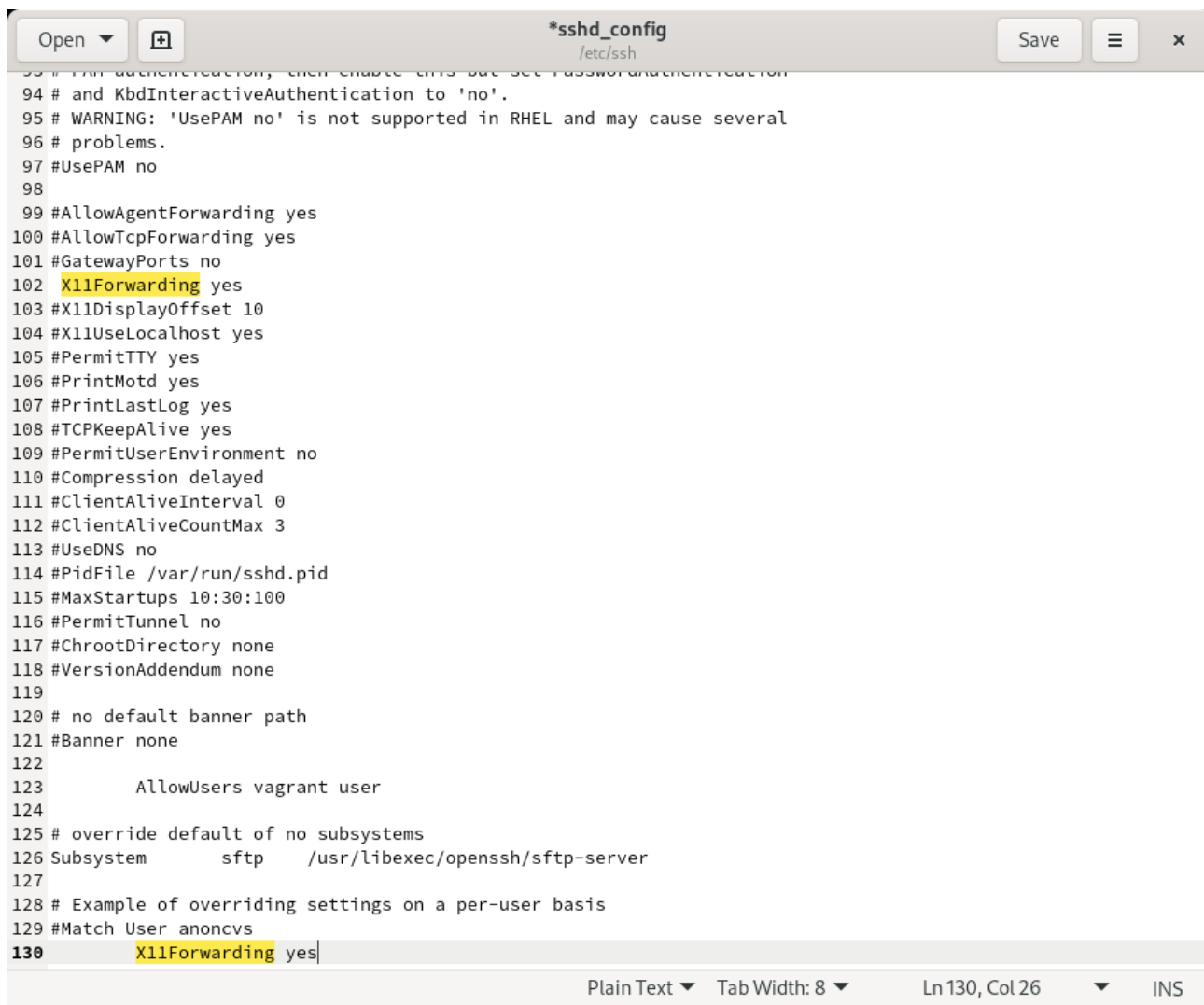
Спасибо за внимание!

Вопросы?



```
root@server:~
to execute child process "dbus-launch" (No such file or directory)
** (gedit:7031): WARNING **: 17:07:31.628: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported
** (gedit:7031): WARNING **: 17:07:31.630: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported
** (gedit:7031): WARNING **: 17:11:37.098: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported
** (gedit:7031): WARNING **: 17:11:37.121: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported
** (gedit:7031): WARNING **: 17:11:38.732: Set document metadata failed: Setting attribute metadata::gedit-position not supported
(gedit:7031): dconf-WARNING **: 17:11:38.761: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server ~]# ssh -fNL 8080:localhost:80 user@server.user.net
The authenticity of host 'server.user.net (fe80::a00:27ff:fe69:a8d%eth0)' can't be established.
ED25519 key fingerprint is SHA256:yiD6CwwpAVWgmZ0LNexnVt25Q77Gy5XU7NfNuqLLz68.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: server
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server.user.net' (ED25519) to the list of known hosts.
user@server.user.net's password:
```

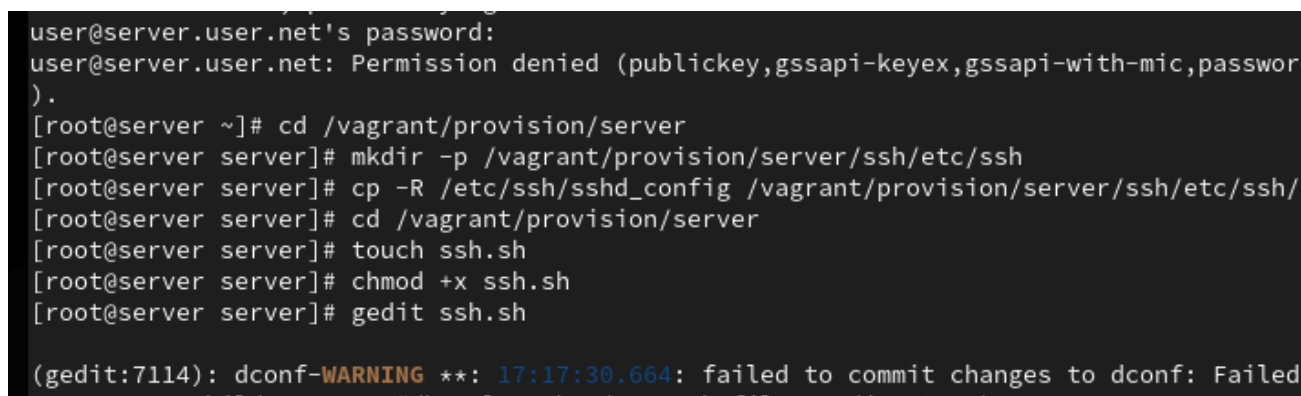
Рис. 12: Этап 12



```
*sshd_config
/etc/ssh

94 # and KbdInteractiveAuthentication to 'no'.
95 # WARNING: 'UsePAM no' is not supported in RHEL and may cause several
96 # problems.
97 #UsePAM no
98
99 #AllowAgentForwarding yes
100 #AllowTcpForwarding yes
101 #GatewayPorts no
102 X11Forwarding yes
103 #X11DisplayOffset 10
104 #X11UseLocalhost yes
105 #PermitTTY yes
106 #PrintMotd yes
107 #PrintLastLog yes
108 #TCPKeepAlive yes
109 #PermitUserEnvironment no
110 #Compression delayed
111 #ClientAliveInterval 0
112 #ClientAliveCountMax 3
113 #UseDNS no
114 #PidFile /var/run/sshd.pid
115 #MaxStartups 10:30:100
116 #PermitTunnel no
117 #ChrootDirectory none
118 #VersionAddendum none
119
120 # no default banner path
121 #Banner none
122
123     AllowUsers vagrant user
124
125 # override default of no subsystems
126 Subsystem      sftp    /usr/libexec/openssh/sftp-server
127
128 # Example of overriding settings on a per-user basis
129 #Match User anoncvs
130     X11Forwarding yes
```

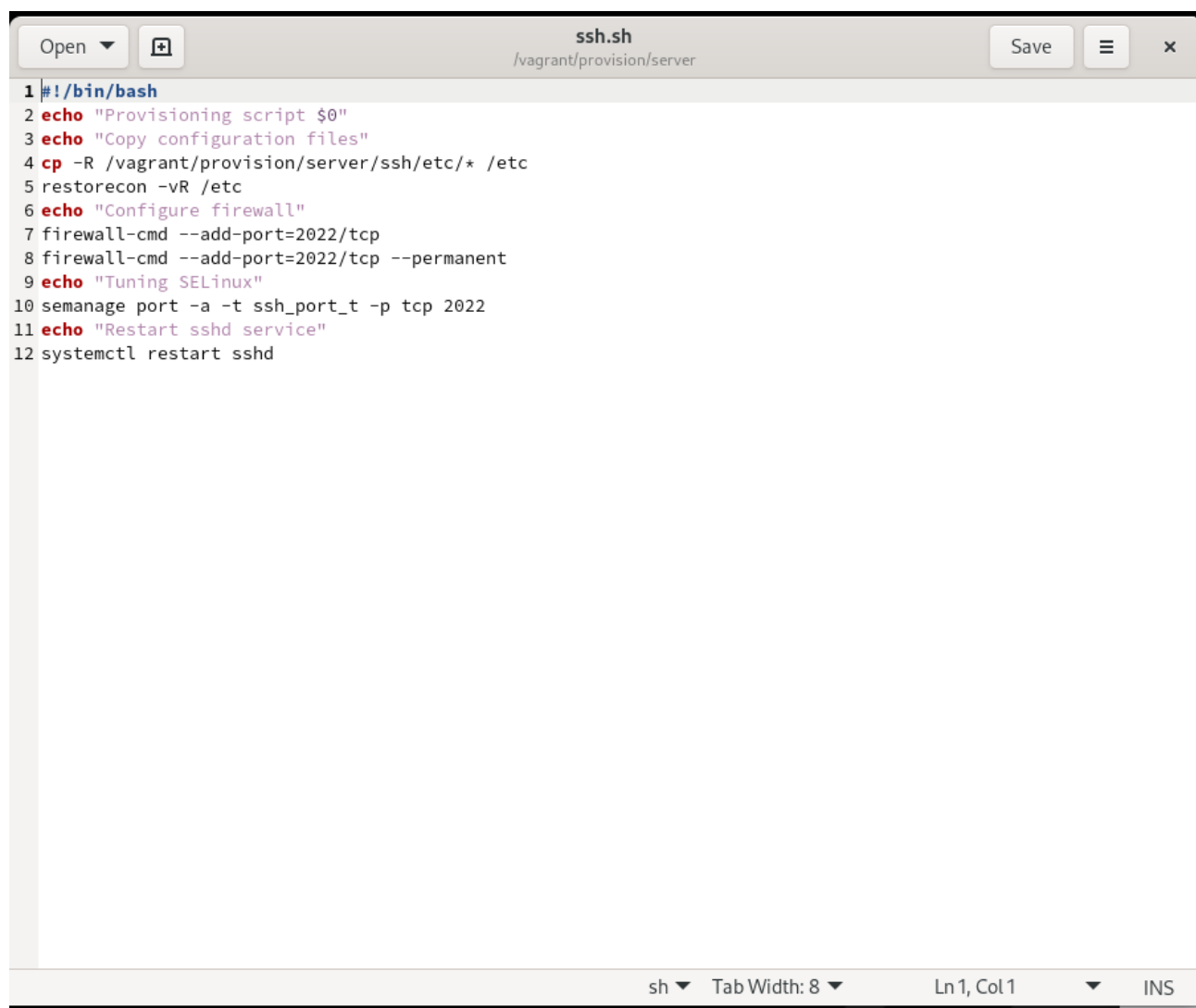
Рис. 13: Этап 13



```
user@server.user.net's password:
user@server.user.net: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
[root@server ~]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/ssh/etc/ssh
[root@server server]# cp -R /etc/ssh/sshd_config /vagrant/provision/server/ssh/etc/ssh/
[root@server server]# cd /vagrant/provision/server
[root@server server]# touch ssh.sh
[root@server server]# chmod +x ssh.sh
[root@server server]# gedit ssh.sh

(gedit:7114): dconf-WARNING **: 17:17:30.664: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
```

Рис. 14: Этап 14



```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Copy configuration files"
4 cp -R /vagrant/provision/server/ssh/etc/* /etc
5 restorecon -vR /etc
6 echo "Configure firewall"
7 firewall-cmd --add-port=2022/tcp
8 firewall-cmd --add-port=2022/tcp --permanent
9 echo "Tuning SELinux"
10 semanage port -a -t ssh_port_t -p tcp 2022
11 echo "Restart sshd service"
12 systemctl restart sshd
```

Рис. 15: Этап 15